

Federated Few-Shot Learning for ICU Treatment Feasibility: Disease-Stratified Simulation on MIMIC-IV

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Abstract

Background. Hospital machine learning often cannot pool raw patient rows across sites because of regulation and trust. Federated learning (FL) trains a global model by aggregating locally computed updates [6], while few-shot meta-learning targets data-scarce adaptation [1, 10]. Their combination is underexplored for *tabular* ICU outcomes.

Objective. Quantify the privacy–utility trade-off when training a prototypical few-shot model under FedAvg-style aggregation on a reproducible MIMIC-IV v3.1 cohort, using diagnosis-based nodes as a proxy for institutional heterogeneity.

Methods. We use $n=2,000$ first ICU stays with five disease strata (sepsis, cardiac, respiratory, renal, diabetes). Training patients are partitioned into five nodes by primary ICD-10 group. Local training uses episodic few-shot updates; the server averages model weights each round. We compare **centralized** training (same data, no partition) vs. **federated** training (`exp3_results.csv`).

Results. Centralized few-shot AUROC **0.552** vs. federated **0.547** — absolute gap **0.005** (0.5 percentage points), with similar F1-macro (**0.543** vs. **0.540**).

Conclusions. In this simulation, FL incurs minimal accuracy loss relative to centralized few-shot training, supporting further research on multi-site clinical FL. However, centralized tabular models reach much higher AUROC (≈ 0.74 – 0.75) on the same cohort; FL comparisons should eventually include federated classical baselines and real network effects.

Keywords: federated learning; few-shot learning; MIMIC-IV; ICU; privacy; electronic health records; FedAvg

1 Introduction

1.1 Motivation

Learning predictive models from electronic health records (EHRs) is constrained by privacy law and institutional policy [2, 7]. FL enables collaborative model improvement without sharing raw records [4, 6]. Medical FL case studies include imaging [9] and growing interest in structured clinical data.

1.2 Problem setting

We study **treatment feasibility** prediction from early ICU labs and vitals on MIMIC-IV [3]. When physical multi-site data are unavailable, **disease-stratified nodes** provide a pragmatic simulation of non-independent, non-identically distributed (non-IID) partitions: sepsis, cardiac, respiratory, renal, and diabetes cohorts differ in lab distributions and outcome rates.

1.3 Contributions

- A fully specified **five-node federated few-shot** protocol with fixed splits and open exports (`exp3_results.csv`).
- Empirical **0.005 AUROC gap** between centralized and federated few-shot training in our harness.
- Critical discussion of **simulation limits** (vs. real hospitals) and **ceiling effects** when comparing to centralized tabular SOTA.

1.4 Organization

Section 2 reviews FL and clinical applications. Section 3 formalizes FedAvg. Section 4 describes the few-shot client objective and server aggregation. Sections 5–6 give experiments and results. Sections 7–8 discuss limitations and conclude.

2 Related work

2.1 Federated optimization

FedAvg [6] remains the default baseline: local SGD epochs followed by weighted averaging. Heterogeneity-aware variants (e.g., FedProx [5]) reduce client drift. Surveys summarize threats, privacy tooling, and systems constraints [4].

2.2 Federated learning in medicine

Overview articles argue FL can enable multi-institutional models without data pooling [7]. Empirical studies include federated tumor segmentation [9]. Structured EHR tasks add schema heterogeneity and variable missingness across sites—challenges our simulation only partially captures.

2.3 Few-shot meta-learning

Prototypical networks [10] and MAML [1] underpin our client-side training. Combining FL with meta-learning has been proposed for cross-device personalization; here we use meta-learning as the *task* model class rather than meta-learning across FL rounds.

3 Preliminaries: FedAvg

Let there be m clients (nodes). Client i holds local dataset $\mathcal{D}_i$. The global loss is

$$\mathcal{L}(\theta) = \sum_{i=1}^m \frac{|\mathcal{D}_i|}{\sum_j |\mathcal{D}_j|} \mathcal{L}_i(\theta), \quad (1)$$

where $\mathcal{L}_i$ is the empirical risk on $\mathcal{D}_i$. Each round t : (1) broadcast θ_t ; (2) each client minimizes $\mathcal{L}_i$ starting from θ_t for E local epochs, yielding $\theta_{t+1}^{(i)}$; (3) server update

$$\theta_{t+1} = \sum_{i=1}^m \frac{|\mathcal{D}_i|}{\sum_j |\mathcal{D}_j|} \theta_{t+1}^{(i)}. \quad (2)$$

In our implementation, clients share the same architecture (ETHOS + prototypical head) but see disjoint patient subsets.

4 Methods

4.1 Outcome, features, and cohort

We mirror the thesis benchmark: binary treatment feasibility from MIMIC-IV v3.1 [3], $n=2,000$ stratified stays, 25-D tabular features from first 24 h (15 labs, 7 vitals, 3 engineered), fixed 64/16/20 split. **Nodes** correspond to primary diagnosis groups (five groups).

4.2 Client model

Each client trains a Symptom Token + ETHOS encoder with prototypical few-shot loss [8, 10] on episodic batches drawn *only* from local patients. Hyperparameters match thesis Experiment 3 (learning rate, local epochs, rounds—see repository `experiments/`).

4.3 Centralized baseline

The centralized trainer uses the **union** of all training patients (still no test leakage) with the same architecture and schedule, providing an upper bound within the few-shot model class *without* partition noise.

4.4 Evaluation

We report global test AUROC and F1-macro on the held-out 400 patients for both centralized and federated global models after training completes.

Algorithm 1 Federated few-shot training (outline)

```
1: Initialize global  $\theta_0$ 
2: for round  $t = 0, 1, \dots, T - 1$  do
3:   for each client  $i \in \{1, \dots, m\}$  in parallel do
4:      $\theta^{(i)} \leftarrow \text{LocalEpisodicTrain}(\theta_t, \mathcal{D}_i, E)$ 
5:   end for
6:    $\theta_{t+1} \leftarrow \sum_i (|\mathcal{D}_i|/|\mathcal{D}|) \theta^{(i)}$ 
7: end for
8: Evaluate  $\theta_T$  on global test set
```

5 Experimental setup

5.1 Software and reproducibility

Python 3.10+, PyTorch, custom FedAvg loop; seed 42; exports in `results/tables/exp3_results.csv`.

5.2 Threat model (informal)

We assume honest-but-curious servers and clients; no differential privacy noise is added in this experiment. Secure aggregation and encryption are important production extensions [4].

5.3 Baselines beyond Exp3

For context, centralized **tabular** models reach AUROC ≈ 0.74 – 0.75 on the same cohort (companion paper / thesis). FL research should report both *within few-shot* and *vs. clinical ceiling* perspectives.

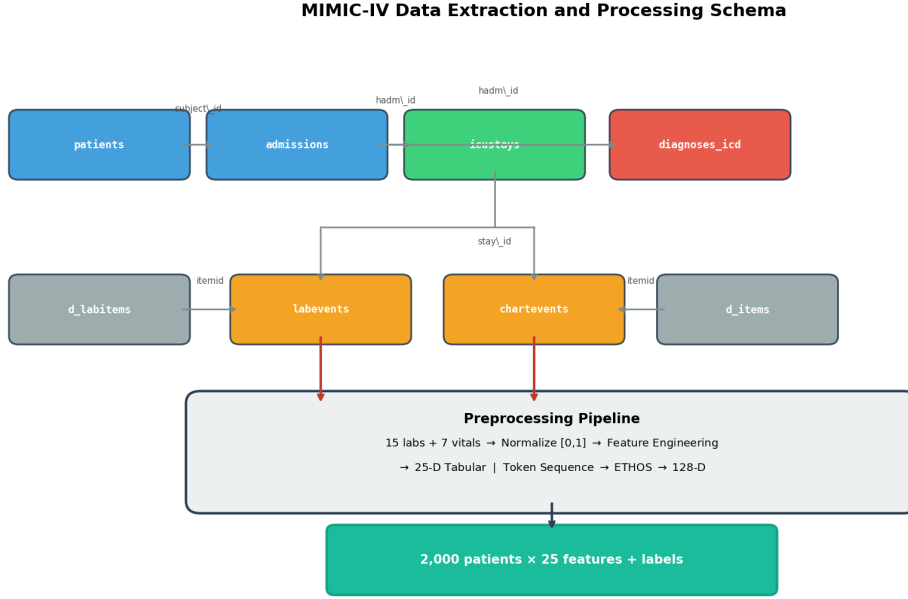


Figure 1: Data pipeline for cohort construction (shared with companion manuscripts).

6 Results

Table 1 summarizes `exp3_results.csv`. The federated global model loses **0.005** AUROC relative to centralized few-shot training.

Table 1: Centralized vs. federated few-shot (Experiment 3, repository export).

Setting	AUROC	F1 (macro)	Rel. to centralized
Centralized	0.552	0.543	—
Federated (5 nodes)	0.547	0.540	−0.005 AUROC

6.1 Interpretation

The small gap suggests that, for this architecture and partition, FedAvg does not catastrophically harm few-shot training. However, **absolute** performance remains far below tuned Random Forest (**0.762** AUROC in Exp1), so FL should be viewed as a **training modality** comparison, not a claim of clinical optimality.

6.2 Node imbalance

Stratified sampling balances nodes at cohort creation; real hospitals would exhibit uneven volumes. Extending the simulation with power-law client sizes is future work.

6.3 Context: centralized tabular and stacked hybrids (same cohort)

For the *same* MIMIC-IV benchmark, centralized **tabular** models reach substantially higher AUROC than the few-shot models used in Exp3. Table 2 (repository exports; see companion manuscript for full analysis) prevents misreading “small FL gap” as “clinical-grade accuracy.”

Table 2: Illustrative centralized ceilings vs. Exp3 few-shot (same cohort, different model classes).

Pipeline	Test AUROC
Best tabular export (<code>exp_final_best_model</code>)	0.747
Tuned RandomForest (<code>exp_tabular_sota</code>)	0.744
Stacked few-shot + classical (<code>exp_fewshot_best</code>)	0.746
Exp3 centralized few-shot	0.552
Exp3 federated few-shot	0.547

7 Discussion

7.1 Simulation fidelity

Diagnosis nodes induce meaningful covariate shift but omit legal data silos, EHR schema mismatch, and asynchronous participation [5]. Results should be cited as **proof-of-concept** for FL optimization stability, not for multi-site generalization.

7.2 Privacy

Update vectors can leak information [4]; DP-FedAvg and secure aggregation are standard mitigations not evaluated here.

7.3 Clinical implications

Low absolute AUROC in the few-shot federated setting implies deployment would be inappropriate without architectural changes (e.g., federated training of *tabular* models) or much larger effective sample sizes.

7.4 Relation to companion work

A companion manuscript details few-shot engineering and stacking that reach ≈ 0.75 AUROC *centrally*; extending those recipes to FL (and comparing to federated XGBoost) is an open engineering task.

8 Conclusion

We evaluated FedAvg-style federated training of a prototypical few-shot ICU feasibility model on a five-node MIMIC-IV simulation. The AUROC drop vs. centralized training is **0.005**, supporting the hypothesis that privacy-preserving aggregation can be cheap in this controlled setting. Real-world FL for ICU prediction will require stronger baselines, security mechanisms, and prospective validation.

Data/code: PhysioNet MIMIC-IV; thesis repository for scripts and CSVs.

Author contributions

M.S. implemented federated simulation and experiments; Dr. A. supervised.

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Competing interests

None declared.

Data and code availability

MIMIC-IV (PhysioNet); code in companion repository.

A Node definitions

Nodes follow primary ICD-10 groups: (1) sepsis (e.g., A40–A41, R65); (2) cardiac (e.g., I21, I50); (3) respiratory (e.g., J18, J44); (4) renal (N17–N19); (5) diabetes (E10–E13). Exact mapping tables are in thesis `extraction/` and appendices.

B Communication cost sketch

Let model size be $|\theta|$ parameters, T rounds. Naive FedAvg uploads/downloads $\mathcal{O}(T \cdot |\theta|)$ per client; compression and partial updates can reduce bandwidth [4].

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